Executive MBA

Financial Accounting

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Lecture 2: Revenue Recognition Topics & Impairment of Receivables

CHICAGO BOOTH

The University of Chicago Booth School of Business

Agenda

- Revenue Recognition Topics:
  - Multidimensional contracts
  - Long-term contracts
  - Contracts with variable consideration
  - Contracts with right of return
  - Principal-agent Contracts
- Impairment of Receivables and Provisioning for Credit Losses

Excerpt from Microsoft's 2024 Annual Report

Revenue Recognition

Our contracts with customers often include promises to transfer multiple products and services to a customer. Determining whether products and services are considered distinct performance obligations that should be accounted for separately versus together may require significant judgment. When a cloud-based service includes both on-premises software licenses and cloud services, judgment is required to determine whether the software license is considered distinct and accounted for separately, or not distinct and accounted for together with the cloud service and recognized over time. Certain cloud services, primarily Office 365, depend on a significant level of integration, interdependency, and interrelation between the desktop applications and cloud services, and are accounted for together as one performance obligation. Revenue from Office 365 is recognized ratably over the period in which the cloud services are provided.

Judgment is required to determine the stand-alone selling price ("SSP") for each distinct performance obligation. We use a single amount to estimate SSP for items that are not sold separately, including on-premises licenses sold with SA or software updates provided at no additional charge. We use a range of amounts to estimate SSP when we sell each of the products and services separately and need to determine whether there is a discount to be allocated based on the relative SSP of the various products and services.

Our products are generally sold with a right of return, we may provide other credits or incentives, and in certain instances we estimate customer usage of our products and services, which are accounted for as variable consideration when determining the amount of revenue to recognize. Returns and credits are estimated at contract inception and updated at the end of each reporting period if additional information becomes available. Changes to our estimated variable consideration were not material for the periods presented.

Key Provisions of Revenue Recognition Standard

- Core principle: “An entity recognizes revenue to depict the transfer of promised goods or services to customers in an amount that reflects the consideration to which the entity *expects* to be entitled in exchange for these goods and services” (emphasis added).

- Application of core principle involves five steps:

  1. Identify the contract with the customer. Contract may be written, oral, or even implied.
  2. Identify the performance obligation(s) in the contract.
  3. Determine the transaction price.
  4. Allocate the transaction price to each performance obligation.
  5. Recognize revenue *when or as* the entity satisfies a performance obligation.

The Five-Step Process—Boeing Example

- Assume that Boeing Corporation signs a contract to sell airplanes to Delta Air Lines for $100 million.

Step 1: Identify the contract with customers.

A contract is an agreement between two parties that creates enforceable rights or obligations. In this case, Boeing has signed a contract to deliver airplanes to Delta.

Step 2: Identify the separate performance obligations in the contract.

Boeing has only one performance obligation—to deliver airplanes to Delta. If Boeing also agreed to maintain the planes, a separate performance obligation is recorded for this promise.

The Five-Step Process—Boeing Example

Step 3: Determine the transaction price.	Transaction price is the amount of consideration that a company expects to receive from a customer in exchange for transferring a good or service. In this case, the transaction price is straightforward—it is $100 million.
Step 4: Allocate the transaction price to the separate performance obligations.	In this case, Boeing has only one performance obligation—to deliver airplanes to Delta.
Step 5: Recognize revenue when each performance obligation is satisfied.	Boeing recognizes revenue of $100 million for the sale of the airplanes to Delta when it satisfies its performance obligation—the delivery of the airplanes to Delta.

The Five Step Process—Oracle Example

In March 2024, Oracle Corp. enters into a contract with a corporate client to transfer database software, perform installation service, and provide *both* unspecified software updates and technical support for a five-year period. Oracle charged the client $1,000,000 for the contract. Oracle sells the database software, installation service, software upgrades, and technical support separately and charges the following amounts, if each element were sold separately:

	Fair Value
Database Management Software	$ 600,000
Installation Service	450,000
Software Upgrades	300,000
Technical support	150,000
Total estimated fair values	$1,500,000

The Five Step Process—Oracle Example

1. Suppose the database software is installed during March 2024 and Oracle receives cash of $1,000,000 from the customer. How much revenue should Oracle recognize during March 2024?

2. How much additional revenue, if any, should Oracle recognize from March 31, 2024, until December 31, 2024, the end of Oracle's fiscal year?

Long-Term Contracts

- Booth Construction signs a contract with Dodo Refinery during fiscal 2024, to build a refinery:
  - ➢ The refinery is built to Dodo Refinery's specifications and the latter can make changes to these specifications over the contract term.
  - ➢ Dodo Refinery can cancel the contract at any time (with a termination penalty), in which case any work-in-process would become the company's property.
  - ➢ The negotiated contract price is $4,500,000 at an estimated cost of $4,000,000.
  - ➢ The following data pertain to the construction period:

	Actual Experience on a Project		
	Fiscal 2024	**Fiscal 2025**	**Fiscal 2026**
Costs incurred to date	$1,000,000	$2,916,000	$4,050,000
Estimated future costs	3,000,000	1,134,000	–
Progress billings to date	900,000	2,400,000	4,500,000
Cash collections to date	750,000	1,750,000	4,500,000

Revenue Recognition: Long-Term Contract

- Suppose the costs incurred each year provide reasonable measures of progress, how should Booth Construction recognize revenues and expenses over the life of the contract?

Fiscal Year	Revenue	Expense	Gross Profit
2024			
2025			
2026			
Cumulative			

Recognizing Revenue or Gross Profit: Percentage-of-Completion Method

$$\frac{\text{Costs incurred to date}}{\text{Most recent estimate of total costs}} = \text{Percent complete}$$

$$\text{Percent complete} \times \begin{matrix}\text{Estimated}\\ \text{total revenue}\\ \text{(or gross profit)}\end{matrix} = \begin{matrix}\text{Revenue (or gross}\\ \text{profit) to be}\\ \text{recognized to date}\end{matrix}$$

$$\begin{matrix}\text{Revenue (or gross}\\ \text{profit) to be}\\ \text{recognized to date}\end{matrix} - \begin{matrix}\text{Revenue (or gross}\\ \text{profit) recognized}\\ \text{in prior periods}\end{matrix} = \begin{matrix}\text{Current-period}\\ \text{revenue}\\ \text{(or gross profit)}\end{matrix}$$

Using Cost Incurred as Measure of Transfer of Control

	Fiscal 2024	Fiscal 2025	Fiscal 2026
Contract price	$4,500,000	$4,500,000	$ 4,500,000
Less estimated costs:			
Cumulative costs incurred	1,000,000	2,916,000	4,050,000
+ Estimated costs to complete	3,000,000	1,134,000	—
= Estimated total costs	4,000,000	4,050,000	4,050,000
Percent complete	$\frac{\$1,000,000}{\$4,000,000} = 25\%$	$\frac{\$2,916,000}{\$4,050,000} = 72\%$	$\frac{\$4,050,000}{\$4,050,000} = 100\%$
Cumulative revenues recognized	0.25 * $4,500,000 = $1,125,000	0.72 * $4,500,000 = $3,240,000	1*$4,500,000
Revenues recognized during fiscal year	$1,125,000	$2,115,000	$1,260,000

Revenue Recognition: Long-Term Contract

- Suppose the costs incurred each year provide a reasonable measure of progress, how should Booth Construction recognize revenues and expenses over the life of the contract?

Fiscal Year	Revenue	Expense	Gross Profit
2024	$1,125,000	1,000,000	$125,000
2025	2,115,000	1,916,000	199,000
2026	1,260,000	1,134,000	126,000
Cumulative	$4,500,000	$4,050,000	$450,000

Long-Term Contracts, cont'd

- But what if the contract had, instead, *at least* one of the following characteristics?

  - Dodo Refinery can cancel the contract at any time (with a termination penalty), and any work-in-process remains the property of Booth Construction.

  - Should the contract be cancelled, the work-in-process can be completed, and Booth Construction can sell the refinery to another customer.

  - Physical possession and title do not pass to Dodo Refinery until completion of the contract.

Long-Term Contracts, cont'd

- Given the modified contract, how should Booth Construction recognize revenues and expenses?

Fiscal	Revenue	Expense	Gross Profit
2024	$0	0	$0
2025	0	0	0
2026	$4,500,000	$4,050,000	$450,000
Total	$4,500,000	$4,050,000	$450,000

Boeing's Long-Term Contracts

For each long-term contract, we determine the transaction price based on the consideration expected to be received. We allocate the transaction price to each distinct performance obligation to deliver a good or service, or a collection of goods and/or services, based on the relative standalone selling prices. A long-term contract will typically represent a single distinct performance obligation due to the highly interdependent and interrelated nature of the underlying goods and/or services and the significant service of integration that we provide. While the scope and price on certain long-term contracts may be modified over their life, the transaction price is based on current rights and obligations under the contract and does not include potential modifications until they are agreed upon with the customer. When applicable, a cumulative adjustment or separate recognition for the additional scope and price may result. Long-term contracts can be negotiated with a fixed price or a price in which we are reimbursed for costs incurred plus an agreed upon profit. The Federal Acquisition Regulations provide guidance on the types of cost that will be reimbursed in establishing the price for contracts with the U.S. government. Certain long-term contracts include in the transaction price variable consideration, such as incentive and award fees, if specified targets are achieved. The amount included in the transaction price represents the expected value, based on a weighted probability, or the most likely amount.

Long-term contract revenue is recognized over the contract term (over time) as the work progresses, either as products are produced or as services are rendered. We generally recognize revenue over time as we perform on long-term contracts because of continuous transfer of control to the customer. For U.S. government contracts, this continuous transfer of control to the customer is supported by clauses in the contract that allow the customer to unilaterally terminate the contract for convenience, pay us for costs incurred plus a reasonable profit and take control of any work in process. Similarly, for non-U.S. government contracts, the customer typically controls the work in process as evidenced either by contractual termination

Boeing's Long-Term Contracts

The accounting for long-term contracts involves a judgmental process of estimating total sales, costs and profit for each performance obligation. Cost of sales is recognized as incurred. The amount reported as revenues is determined by adding a proportionate amount of the estimated profit to the amount reported as cost of sales. Recognizing revenue as costs are incurred provides an objective measure of progress on the long-term contract and thereby best depicts the extent of transfer of control to the customer.

Changes in estimated revenues, cost of sales and the related effect on operating income are recognized using a cumulative catch-up adjustment which recognizes in the current period the cumulative effect of the changes on current and prior periods based on a long-term contract's percentage-of-completion. When the current estimates of total sales and costs for a long-term contract indicate a loss, a provision for the entire reach-forward loss on the long-term contract is recognized.

Contracts with Variable Consideration

- In practice, many contracts contain an element of consideration that is variable or contingent upon certain thresholds or events being met or achieved, such as returns and volume discounts (manufacturing), incentive payments (construction), rebates (telecommunications), and royalties (entertainment and media).

- Should variable consideration be included in the transaction price?
  - Yes…but only to the extent that it is highly probable (in practice, this means a probability greater than 50%) that a significant reversal in the amount of cumulative revenue recognized will not occur when the uncertainty associated with the variable consideration is subsequently resolved.
  - If the likelihood of receiving the variable consideration changes over the life of the contract, then the amount of variable consideration recognized should be adjusted.

- If variable consideration is included, what should be the amount?
  - Expected amount based on a weighted probability.
  - Most likely amount used in the case of binary outcomes.

Expected Value vs. Most Likely Amount

- **Expected Value**: Probability-weighted amount in a range of possible consideration amounts.
  - ➢ May be appropriate if a company has a large number of contracts with similar characteristics.
  - ➢ Can be based on a limited number of discrete outcomes and probabilities.
- **Most Likely Amount**: The single most likely amount in a range of possible consideration outcomes.
  - ➢ May be appropriate if the contract has only two possible outcomes.

Contracts with Variable Consideration: An Example

On July 1, 2024, BoothTech enters into a contract with Thaler Gambling (Thaler) to feature Thaler's online game *Dare to Nudge* on BoothTech's network. Thaler pays BoothTech an upfront fee of $300,000 in cash for six months of featured access. Thaler will also pay BoothTech a bonus of $180,000 if BoothTech's users access Thaler's online games for at least 15,000 hours during the fiscal year 2024 ending on 31/12/2024.

- On July 1, 2024, BoothTech estimates that there is a 75% chance that it will achieve the usage target and receive the $180,000 bonus at the end of fiscal 2024.

- On October 1, 2024, BoothTech concludes that it was overconfident in its initial assessment of the likelihood of receiving the bonus. It now believes it is unlikely that it will receive the bonus.

- As of December 31, 2024, BoothTech's users have accessed Thaler's online game for a total of 6,500 hours.

How should BoothTech recognize revenues during fiscal 2024? Assume BoothTech measures variable consideration at its most likely amount.

Contracts with Variable Consideration: An Example

July 1: Expected Transaction Price =

July 31, August 31, September 30:

October 1: Revised Expected Transaction Price =

October 31, November 30. and December 31:

Contracts with Right of Return

Consider a firm that sells a line of its products for $840,000 in cash during fiscal 2024. The products have a gross margin of 40%. Customers have a right of return that expires at the end of fiscal 2024. During fiscal 2024, customers return products with a total sales price of $24,000. The inventory products are returned in their original condition.

A. Suppose, based on past experience, the firm expects 10% of the products might be returned during fiscal 2024 (i.e., the firm estimates that it is probable that 10% of the products would be returned).

B. Suppose, instead, the firm's line of products is new, so it has no historical experience with the product. Consequently, the firm cannot reliably estimate the amount of product returns.

A. Cash Sales: Firm *can* reliably estimate returns

- During fiscal 2024, firm recognizes:

Cash	\$840,000	
Revenue-Gross		\$840,000
Revenue Returns (contra revenue)	\$84,000	
Liability for Returns (Liability)		\$84,000

- Note that the journal entries could be combined so that revenues are recognized at the *expected transaction price* of \$840,000*0.9 = \$756,000.

Cash	\$840,000	
Revenue-net of returns		\$756,000
Liability for Returns		84,000
COGS (\$756,000*0.6)	\$453,600	
Inventory with customer	50,400	
	Inventory	\$504,000

A. Cash Sales: Firm *can* reliably estimate returns

- To recognize actual returns during fiscal 2024:

Liability for Returns	$24,000	
Cash		$24,000

Inventory	$14,400	
Inventory with customer		$14,400

- At the end of fiscal 2024, right of return expires and if there are no other products with right of return policies, any remaining balance in the liability and sales returns accounts are eliminated and revenues and COGS associated with the outstanding products are recognized:

Liability for Returns	$60,000	
Revenue Returns		$60,000

COGS	$36,000	
Inventory with customer		$36,000

A. Cash Sales : Firm *can* reliably estimate returns

- Suppose, instead, goods worth $24,000 were returned, but the goods were spoiled and had to be disposed of.

- Suppose, instead, goods worth $100,000 (instead of $24,000) were returned during fiscal 2024?

A. Cash Sales: Firm *cannot* reliably estimate returns

- If the firm *cannot* reliably estimate the amount of product returns, then during fiscal 2024, the following transactions are recognized:

Cash	$840,000	
Liability for Returns		$840,000
Liability for Returns	$24,000	
Cash		$24,000
Inventory with customer	$489,600	
Inventory		$489,600

- After right of return expires at the end of fiscal 2024:

Liability for Returns	$816,000	
Revenues		$816,000
COGS	$489,600	
Inventory with customer		489,600

A. Sales on credit and firm *can* estimate returns

Suppose, based on past experience, the firm estimates 10% of products could be returned during the year.

Account Receivable-Gross	$840,000 (given)	
Revenue-Gross		$840,000
Revenue Returns (contra revenue)	$84,000	
Allowance for Returns (contra-asset)		84,000

The two journal entries above could be combined so that revenues are recognized at the *expected transaction price* of $840,000*0.9 = $756,000.

Account Receivable, net of returns	$756,000	
Revenue-net of returns		$756,000
COGS ($756,000*0.6)	$453,600	
Inventory with customer	50,400	
	Inventory	$504,000

A. Sales on credit and firm *can* reliably estimate returns

- To recognize actual returns during fiscal 2024

Allowance for Returns	$24,000	
Accounts Receivable-Gross		$24,000

Inventory	$14,400	
Inventory with customer		$14,400

- Note that, for both cash and credit sales, the *actual* level of product returns has no impact on revenues or COGS since the firm has *already over-provisioned* for returns during the year. Similarly, actual returns have no impact on A/R–net and only reduce A/R–Gross.

- After right of return expires at the end of fiscal 2024 and there are no other product lines with right of return policies:

Allowance for Returns	$60,000	
Revenue-Returns		$60,000
COGS	$36,000	
Inventory with customer		36,600

A. Sales on credit and firm can reliably estimate returns

- But what if actual returns exceed the total amount provisioned, i.e., the actual returns were \$100,000 so that the firm *under-provisioned* for returns during fiscal 2024?

Allowance for Returns	\$84,000	
Revenues	16,000	
	Account Receivable-Gross	\$100,000
Inventory	\$60,000	
	Inventory with customer	\$50,400
	COGS	9,600

B. Sales on credit but firm *cannot* reliably estimate returns

During fiscal 2024, the firm cannot recognize any revenues *until* the right of return expires at the end of fiscal 2024. Assuming only goods worth $24,000 are returned during fiscal 2024, then at the end of fiscal 2024, the following entries are made:

Accounts Receivable–Gross/Cash	$816,000	
Revenue		$816,000
COGS	$489,600	
Inventory		489,600

Principal-Agent Contracts

- Many contracts are structured in such a way that an agent's obligation is to arrange for the principal to provide goods or services to a customer; e.g., Amazon, Alibaba, eBay, Airbnb, or Uber.

- For such contracts, how do we distinguish between the agent and the principal?
  - Who bears inventory risk?
  - Who controls prices of products/services?
  - Who bears credit risk of customer?

- Why is the distinction important?
  - It affects the amount of revenue that a company can record.
  - If **principal**: recognize revenue on *gross* basis = value of goods/services provided as well as COGS = cost of goods/services provided.
  - If **agent**: recognize revenue on a *net* basis = commissions received on transaction.

Principal-Agent Contracts: An Example

- Manufacturer ships inventory of its latest products costing $1,000,000 to a retailer during fiscal 2025. Retailer pays local advertising costs of $20,000 that are reimbursable by Manufacturer. By the end of fiscal 2025, Retailer sells half of the inventory for $575,000 in cash to customers. Retailer notifies Manufacturer of the sale, retains a 5% sales commission, and remits the cash due to Manufacturer.

  - ➢ Assume that the terms of the contract are such that the manufacturer bears most of the inventory and credit risks: How and when should the manufacturer and retailer recognize revenues during fiscal 2025?

Consignment Sales

Manufacturer		Retailer	
Shipment of Merchandise			
Inventory on Consignment	$1,000,000	No entry	
Inventory	1,000,000		
Alternatively, since inventory is just relabeled on Manufacturer's books, no entry required on the date the goods are shipped.			
Payment of Advertising by Retailer			
No entry until notified		Receivable from Manufacturer	$20,000
		Cash	$20,000
Sale by Retailer			
No entry until notified		Cash	$575,000
		Payable to Manufacturer	$575,000

Consignment Sales

Manufacturer			Retailer		
Notification of sales and expenses and remittance of amount due					
Cash	$526,250		Payable to Manufacturer	$575,000	
Advertising Expense	20,000		Receivable from Manufacturer		20,000
Commission Expense	28,750		Commission Revenue		28,750
Sales		575,000	Cash		526,250
COGS	$500,000				
Inventory on Consignment		500,000			
[1/2 * $1,000,000]					
If inventory is not reclassified at shipment, then the following entry should be made:					
COGS	$500,000				
Inventory		500,000			

Impairment of Receivables

There are two basic methods for measuring the impairment of receivables

1. Direct Write-off Method
   (not allowed under US GAAP / IFRS, but allowed for tax purposes in many jurisdictions)

OR

2. Allowance or Provision Method

The Direct Write-off Method

- Any increase in the risk of non-collection is not recognized until the receivable is written off. In other words, no impairment loss on the receivable is recognized until the account is written off.
  - This method is typically required by tax jurisdictions.

- Writing off Receivables worth *$X*:

Impairment Loss or Bad Debt Expense	$X	
Receivables		$X

- Shortcomings of the direct write-off method:
  - Distorts net income by delaying recognition of impairment loss until the account is determined to be uncollectible.
  - Does not provide the best estimate of how receivables affect expected future cash inflows since receivables may be overstated.

The Allowance or Provision Method

- Under this method, any increase in the risk of non-collection (credit risk) is measured and recognized every period using an *estimation model*.
  - ➢ An impairment loss on the receivable is recognized *even though* the customer has not defaulted and may never default!
  - ➢ Note that, since the receivable is impaired but not written off, an allowance account is used to write down the value of the receivable.
- US GAAP and IFRS require that “bad debt expense” (term used for impairment of trade receivables) or “provision for credit loss” (terms used for financial instruments such as loans) be estimated and recognized as a component of an entity’s SG&A as follows:

Bad Debt Expense/Provision for Credit Loss	$X	
Allowance for Uncollectibles / Loan Losses		$X

The Allowance or Provision Method

- The "Allowance for Uncollectibles" or "Allowance for Loan Losses" account is a ***contra asset*** account and hence has a normal credit balance:
  - When it is subtracted from *Receivables/Loans*, the difference is called *Receivables/Loans–Net*, which estimates the net realizable value of receivables/loans.
- Under the allowance or provision method, when a specific account receivable worth *$Y* from a customer is known to be uncollectible, the entire account must be written off from the books as follows:

Allowance for Uncollectibles	$Y	
Accounts Receivable–Gross		$Y

- Note that, unlike the direct write-off method, the *actual* write-off of an account under the Allowance Method has no impact on bad debt expense as long as the firm has *adequately provisioned* for write-offs during the year. Similarly, actual write-offs have no impact on A/R–Net and only reduce A/R–Gross.

Entries under the Allowance Method during Fiscal Year

(1) Recognizing credit sales worth $x		
Accounts Receivable–Gross	$x	
Revenues-Gross		$x

(2) Cash $y collected from customers for prior credit sales		
Cash	$y	
Accounts Receivable-Gross		$y

(3) Net write-offs/charge-offs of specific accounts receivable worth $z		
Allowance for Uncollectibles (B/S)	$z	
Accounts Receivable		$z

(4) Provisioning for impairment using an estimation model (usually an adjusting entry made at the end of a firm's fiscal year):		
Bad debt expense (I/S)	$b	
Allowance for Uncollectibles (B/S)		$b

Estimating Impairment Loss in Practice…

Allowance for Loan Losses

	Beginning Balance
(*actual*) Net Charge-offs	Provision for credit loss *(derived)*
	Ending Balance *(estimated)*

- The ending balance is typically estimated to reflect the amount *not* expected to be collected using an estimation model:
  - Under current GAAP, the model used to estimate the ending balance is called an *expected loss model*, which requires management to estimate expected credit losses over the life of the receivable. Impairment loss is then calculated as the amount necessary to adjust the allowance to that desired balance.

JP Morgan Chase

JPMorgan Chase & Co.
Consolidated balance sheets

December 31, (in millions, except share data)	2022	2021
Assets		
Cash and due from banks	**$ 27,697**	$ 26,438
Deposits with banks	**539,537**	714,396
Federal funds sold and securities purchased under resale agreements (included **$311,883** and $252,720 at fair value)	**315,592**	261,698
Securities borrowed (included **$70,041** and $81,463 at fair value)	**185,369**	206,071
Trading assets (included assets pledged of **$93,687** and $102,710)	**453,799**	433,575
Available-for-sale securities (amortized cost of **$216,188** and $308,254; included assets pledged of **$9,158** and $18,268)	**205,857**	308,525
Held-to-maturity securities	**425,305**	363,707
Investment securities, net of allowance for credit losses	**631,162**	672,232
Loans (included **$42,079** and $58,820 at fair value)	**1,135,647**	1,077,714
Allowance for loan losses	**(19,726)**	(16,386)
Loans, net of allowance for loan losses	**1,115,921**	1,061,328
Accrued interest and accounts receivable	**125,189**	102,570
Premises and equipment	**27,734**	27,070
Goodwill, MSRs and other intangible assets	**60,859**	56,691
Other assets (included **$14,921** and $14,753 at fair value and assets pledged of **$7,998** and $5,298)	**182,884**	181,498
Total assets[(a)]	**$ 3,665,743**	$ 3,743,567
Liabilities		
Deposits (included **$28,620** and $11,333 at fair value)	**$ 2,340,179**	$ 2,462,303
Federal funds purchased and securities loaned or sold under repurchase agreements (included **$151,999** and $126,435 at fair value)	**202,613**	194,340
Short-term borrowings (included **$15,792** and $20,015 at fair value)	**44,027**	53,594
Trading liabilities	**177,976**	164,693
Accounts payable and other liabilities (included **$7,038** and $5,651 at fair value)	**300,141**	262,755
Beneficial interests issued by consolidated VIEs (included **$5** and $12 at fair value)	**12,610**	10,750
Long-term debt (included **$72,281** and $74,934 at fair value)	**295,865**	301,005
Total liabilities[(a)]	**3,373,411**	3,449,440
Commitments and contingencies (refer to Notes 28, 29 and 30)		
Stockholders' equity		
Preferred stock ($1 par value; authorized 200,000,000 shares: issued **2,740,375** and 3,483,750 shares)	**27,404**	34,838
Common stock ($1 par value; authorized 9,000,000,000 shares; issued **4,104,933,895** shares)	**4,105**	4,105
Additional paid-in capital	**89,044**	88,415
Retained earnings	**296,456**	272,268
Accumulated other comprehensive losses	**(17,341)**	(84)
Treasury stock, at cost (**1,170,676,094** and 1,160,784,750 shares)	**(107,336)**	(105,415)
Total stockholders' equity	**292,332**	294,127
Total liabilities and stockholders' equity	**$ 3,665,743**	$ 3,743,567

JP Morgan Chase

JPMorgan Chase & Co.
Consolidated statements of income

Year ended December 31, (in millions, except per share data)		2022		2021		2020
Revenue						
Investment banking fees	$	**6,686**	$	13,216	$	9,486
Principal transactions		**19,912**		16,304		18,021
Lending- and deposit-related fees		**7,098**		7,032		6,511
Asset management, administration and commissions		**20,677**		21,029		18,177
Investment securities gains/(losses)		**(2,380)**		(345)		802
Mortgage fees and related income		**1,250**		2,170		3,091
Card income		**4,420**		5,102		4,435
Other income		**4,322**		4,830		4,865
Noninterest revenue		**61,985**		69,338		65,388
Interest income		**92,807**		57,864		64,523
Interest expense		**26,097**		5,553		9,960
Net interest income		**66,710**		52,311		54,563
Total net revenue		**128,695**		121,649		119,951
Provision for credit losses		**6,389**		(9,256)		17,480
Noninterest expense						
Compensation expense		**41,636**		38,567		34,988
Occupancy expense		**4,696**		4,516		4,449
Technology, communications and equipment expense		**9,358**		9,941		10,338
Professional and outside services		**10,174**		9,814		8,464
Marketing		**3,911**		3,036		2,476
Other expense		**6,365**		5,469		5,941
Total noninterest expense		**76,140**		71,343		66,656
Income before income tax expense		**46,166**		59,562		35,815
Income tax expense		**8,490**		11,228		6,684
Net income	$	**37,676**	$	48,334	$	29,131
Net income applicable to common stockholders	$	**35,892**	$	46,503	$	27,410
Net income per common share data						
Basic earnings per share	$	**12.10**	$	15.39	$	8.89
Diluted earnings per share		**12.09**		15.36		8.88
Weighted-average basic shares		**2,965.8**		3,021.5		3,082.4
Weighted-average diluted shares		**2,970.0**		3,026.6		3,087.4

JP Morgan Chase

JPMorgan Chase & Co.
Consolidated statements of cash flows

Year ended December 31, (in millions)	2022	2021	2020
Operating activities			
Net income	**$ 37,676**	$ 48,334	$ 29,131
Adjustments to reconcile net income to net cash provided by/(used in) operating activities:			
Provision for credit losses	**6,389**	(9,256)	17,480
Depreciation and amortization	**7,051**	7,932	8,614
Deferred tax (benefit)/expense	**(2,738)**	3,748	(3,573)
Other	**5,174**	3,274	1,649
Originations and purchases of loans held-for-sale	**(149,167)**	(347,864)	(166,504)
Proceeds from sales, securitizations and paydowns of loans held-for-sale	**167,709**	336,413	175,490
Net change in:			
Trading assets	**(31,449)**	85,710	(148,749)
Securities borrowed	**20,203**	(45,635)	(20,734)
Accrued interest and accounts receivable	**(22,970)**	(12,401)	(18,012)
Other assets	**(2,882)**	(11,745)	(42,430)
Trading liabilities	**11,170**	(23,190)	77,198
Accounts payable and other liabilities	**58,614**	43,162	7,415
Other operating adjustments	**2,339**	(398)	3,115
Net cash provided by/(used in) operating activities	**107,119**	78,084	(79,910)

JP Morgan Chase

(Table continued on next page)

Year ended December 31, (in millions)	2022 Consumer, excluding credit card	Credit card	Wholesale	Total
Allowance for loan losses				
Beginning balance at January 1,	$ 1,765	$ 10,250	$ 4,371	$ 16,386
Cumulative effect of a change in accounting principle[(a)]	NA	NA	NA	NA
Gross charge-offs	812	3,192	322	4,326
Gross recoveries collected	(543)	(789)	(141)	(1,473)
Net charge-offs	269	2,403	181	2,853
Provision for loan losses	543	3,353	2,293	6,189
Other	1	–	3	4
Ending balance at December 31,	$ 2,040	$ 11,200	$ 6,486	$ 19,726

Allowance for Loan Losses

	$16,386
(Net charge-offs) $2,853	(Provision for loan losses) 6,189
	(Other) 4
	$19,726

Summary: Revenue Recognition Journal Entries

Journal Entries

(1) *Credit Sale*

Dr A/R-Gross
  Cr Revenue-Gross

(2) *A/R Collection*

Dr Cash
  Cr A/R (Gross)

(3) *Cash Sale*

Dr Cash
  Cr Revenue (Gross)

(4) *Provision for Uncollectibles*

Dr Bad Debt Expense
  Cr Allowance for Uncollectibles (Contra-A/R)

(5) *Provision for Sales Returns (Credit Sales)*

Dr Sales Returns (Contra-Revenue)
  Cr Allowance for Sales Returns

(6) *Provision for Sales Returns (Cash Sales)*

Dr Sales Returns (Contra-Revenue)
  Cr Liability for Sales Returns

(7) *(Actual) Write-off of Uncollectible A/R*

Dr Allowance for Uncollectibles (Contra-A/R)
  Cr A/R (Gross)

(8) *(Actual) Returns of Sold Goods (Credit Sales)*

Dr Allowance for Sales Returns (Contra-A/R)
  Cr A/R (Gross)

(9) *(Actual) Returns of Sold Goods (Cash Sales)*

Dr Liability for Returns
  Cr Cash

(10) *Advances from Customers*

Dr Cash
  Cr Deferred Revenue

(11) *Reversal of Deferred Revenue*

Dr Deferred Revenue
  Cr Revenue (Gross)

Summary: Revenue Recognition T-Accounts

A/R (Gross)

BB			
Credit Sales	(1)	Collections	(2)
		Write-offs	(7)
		Actual Returns	(8)
EB			

Allowance for Uncollectibles

		BB	
Write-offs	(7)	Provision for Uncollectibles	(4)
		EB	

Allowance for Sales Returns

		BB	
Actual Product Returns (Credit Sales)	(8)	Provision for Sales Returns	(5)
		EB	

Revenue (Gross)

		Credit Sales	(1)
		Cash Sales	(3)
		Reversals	(11)

Bad Debt Expense

Provision for Uncollectibles	(4)		

Sales Returns

Provision for Sales Returns	(5)		
Provision for Sales Returns	(6)		

Cash

BB			
Collections	(2)	Actual Returns (Cash Sales)	(9)
Cash Sales	(3)		
Advances	(10)		
EB			

Deferred Revenue

		BB	
Reversals	(11)	Advances	(10)
		EB	

Liability for Returns

		BB	
Actual Returns (Cash Sales)	(9)	Estimated Liability for Sales Returns	(6)
		EB	

Net Accounts:

A/R (Net)

BB_{net}		Collections (2)
Credit Sales	(1)	Provision for Uncollectibles (4)
		Provision for Sales Returns (5)
EB_{net}		

Revenue (Net)

Provision for Sales Returns-Credit Sales	(5)	Credit Sales	(1)
Provision for Sales Returns-Cash Sales	(6)	Cash Sales	(3)
		Reversals of Deferred Revenue	(11)

where $BB_{net} = BB_{gross} - BB_{\text{Allowances for Uncollectibles}} - BB_{\text{Allowances for Sales Returns}}$

and $EB_{net} = EB_{gross} - EB_{\text{Allowances for Uncollectibles}} - EB_{\text{Allowances for Sales Returns}}$